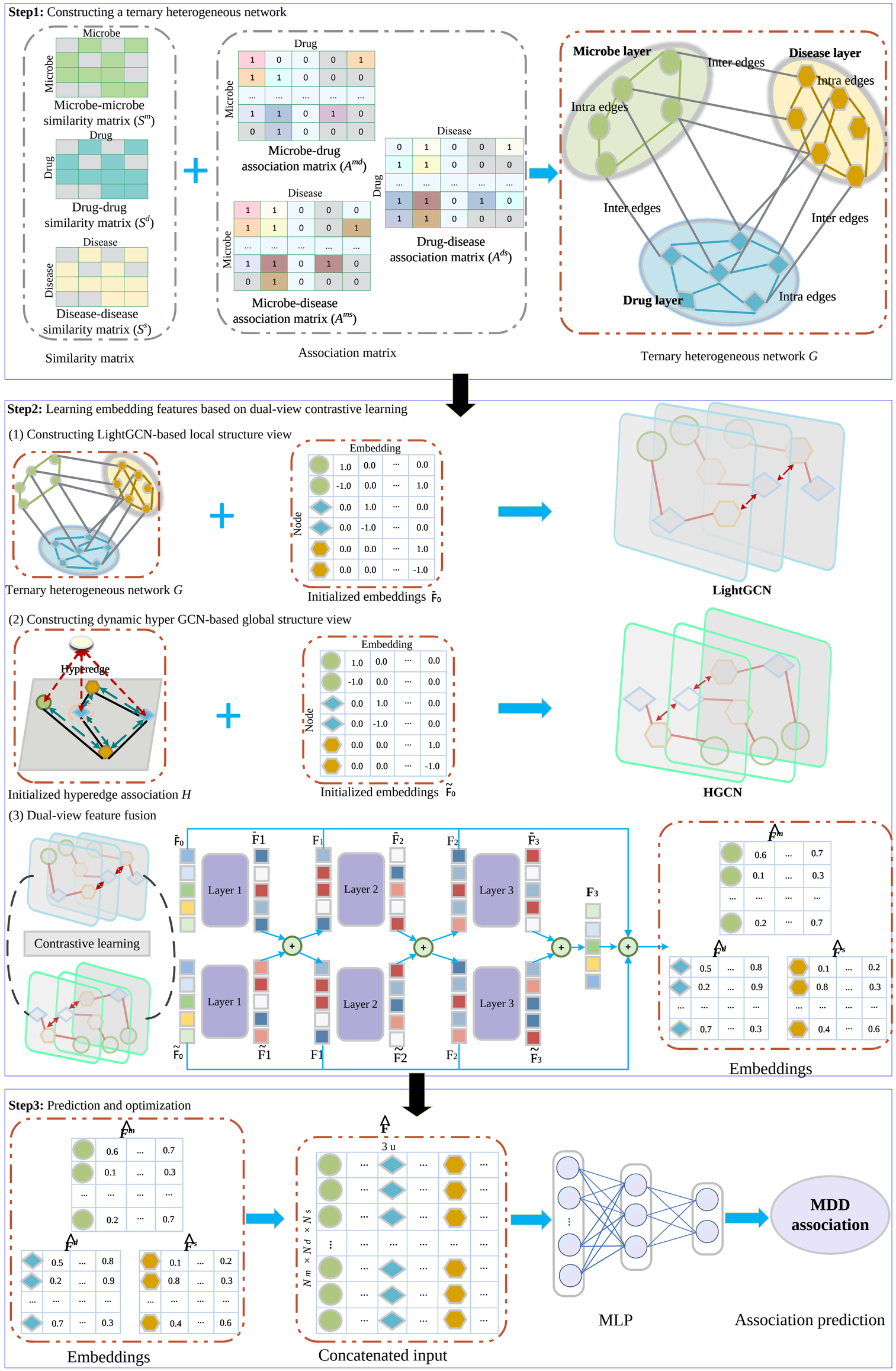
Step1: Constructing a ternary heterogeneous network
Microbe
Microbe-microbe similarity matrix (S^m)
Drug
Drug-drug similarity matrix (S^d)
Disease
Disease-disease similarity matrix (S^s)
Similarity matrix
Microbe-drug association matrix (A^{md})
Drug-disease association matrix (A^{ds})
Microbe-disease association matrix (A^{ms})
Association matrix
Microbe layer
Disease layer
Drug layer
Inter edges
Intra edges
Ternary heterogeneous network G
Step2: Learning embedding features based on dual-view contrastive learning
(1) Constructing LightGCN-based local structure view
Ternary heterogeneous network G
Embedding
Node
Initialized embeddings $\bar{F}_0$
LightGCN
(2) Constructing dynamic hyper GCN-based global structure view
Hyperedge
Initialized hyperedge association H
Initialized embeddings $\tilde{F}_0$
HGCN
(3) Dual-view feature fusion
Contrastive learning
Layer 1
Layer 2
Layer 3
Embeddings
Step3: Prediction and optimization
Embeddings
Concatenated input
MLP
MDD association
Association prediction